Type MS · Trig equations that factorise

Answer 1 of 7 · Question 8 part b · 4 marks

2020 JAN · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

Number		
8 (a)	$2x - 75 = -31, 211$ $x = 22, 143$	M1A1 A1 (3)
(b)	$2\frac{\sin y^\circ}{\cos y^\circ} + 5\sin y^\circ = 0$ $\sin y^\circ\left(\frac{2}{\cos y^\circ} + 5\right) = 0$ $\cos y^\circ = -\frac{2}{5}$ $(\sin y^\circ = 0)$ $y = 113.6^\circ$ $y = 0^\circ, 180^\circ$	M1 M1 A1 B1 (4)
(c)	$3(1 - \sin^2\theta) - 3\sin^2\theta + \sin\theta + 12 = 0$ $6\sin^2\theta - \sin\theta - 15 = 0$ $(2\sin\theta + 3)(3\sin\theta - 5) = 0$ $\sin\theta = -\frac{3}{2}$ $\sin\theta = \frac{5}{3}$ As $-1 \leq \sin\theta \leq 1$ no such values for θ exist	M1 M1 A1 B1 (4)
		[11]

Type MS · Trig equations that factorise

Answer 1 of 7 · Question 8 part b · 4 marks

2020 JAN · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For finding at least one correct value of $(2x-75)=-31^\circ$ or 211° and attempting to find one value of x $\Rightarrow x=\frac{-31+75}{2}$ or $x=\frac{211+75}{2}$
	A1	For $x=22$ or 143
	A1	For $x=22$ and 143 Extra values within range – A0 Extra values outside of the range - ignore
.(b)	M1	For using the identity $\tan y^\circ=\frac{\sin y^\circ}{\cos y^\circ}$
	M1	For factorising their expression and finding values for $\sin y^\circ$ and $\cos y^\circ$ $\sin y^\circ\left(\frac{2}{\cos y^\circ}+5\right)=0\Rightarrow \sin y^\circ=0,\ \cos y^\circ=-\frac{2}{5}\Rightarrow y=...$
	A1	For $y=113.6$ if there are extra values within range – A0
	B1	For both $y=0$ and 180 **Both required**
	ALT	
	M1	For multiplying $\sin y\times\frac{\cos y}{\cos y}\Rightarrow \tan y\cos y\Rightarrow(2\tan y+5\tan y\cos y=0)$
	M1	For factorising the above expression and finding values for $\tan y^\circ$ and $\cos y^\circ$ $2\tan y+5\tan y\cos y=0\Rightarrow \tan y(2+5\cos y)=0$ $\Rightarrow \tan y=0,\ \cos y=-\frac{2}{5}\Rightarrow y=...$
	A1	For $y=113.6$ Extra values within range – A0 Extra values outside of the range - ignore
	B1	For both $y=0$ and 180 **Both required**
	SC	$2\frac{\sin y}{\cos y}=-5\sin y\Rightarrow \cos y=-\frac{2}{5}\Rightarrow y=113.6$ no evidence of factorising – award M1M0A1B0 only (unless there is later recovery)
(c)	M1	For using the identity $\cos^2\theta+\sin^2\theta=1\Rightarrow 3(1-\sin^2\theta)-3\sin^2\theta+\sin\theta+12=0$ to form a 3TQ in terms of $\sin\theta$ Minimally acceptable attempt is $6\sin^2\theta\pm\sin\theta\pm15=0$
	M1	For an attempt to solve their 3TQ (see general guidance for the definition of an attempt $6\sin^2\theta-\sin\theta-15=0\Rightarrow(2\sin\theta+3)(3\sin\theta-5)=0\Rightarrow\sin\theta=...,...$
	A1	$\sin\theta=-\frac{3}{2},\ \frac{5}{3}$
	B1	For the conclusion; $\lvert\sin\theta\rvert>1$ therefore no values exist for $\sin\theta$ Do not accept 'undefined' without an explanation that $\lvert\sin\theta\rvert>1$
Penalise rounding only once in this question		

Type MS · Trig equations that factorise

Answer 1 of 7 · Question 8 part b · 4 marks

2020 JAN · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Trig equations that factorise

Answer 2 of 7 · Question 1 part whole · 5 marks

2025 JUN · Paper 02 · extract 1 of 2 · find part whole in this mark scheme extract

Number		
1	$\sqrt{3}\tan A = 2\sin A \Rightarrow \dfrac{\sqrt{3}\sin A}{\cos A} = 2\sin A$	M1
	$\Rightarrow \sqrt{3}\sin A - 2\sin A\cos A = 0 \Rightarrow \sin A\left(\sqrt{3} - 2\cos A\right) = 0$	M1
	$\Rightarrow \sin A = 0 \quad \text{or} \quad \cos A = \dfrac{\sqrt{3}}{2}$	M1
	$\Rightarrow A = \sin^{-1} 0 = 0^\circ, 180^\circ, 360^\circ \quad , \quad \cos A = \dfrac{\sqrt{3}}{2} \Rightarrow A = 30^\circ, 330^\circ$	A1A1 [5]
		Total 5 marks

Mark	Additional Guidance
M1	Uses identity $\tan = \dfrac{\sin}{\cos}$
M1	Rearranges to obtain $\sqrt{3}\sin A - 2\sin A\cos A = 0$ or it's equivalent **AND** factorises to obtain $\sin A\left(\sqrt{3} - 2\cos A\right) = 0$ This step can be implied by further correct work. If they cancel through by $\sin A$ this is M0
M1	Finds any one value for either $\sin A$ or $\cos A$
A1	For at least any two correct angles obtained from either $\cos A$ or $\sin A$
A1	For all five correct angles Ignore any values outside of the range. Withold this mark for any extra angles within range.

Type MS · Trig equations that factorise

Answer 2 of 7 · Question 1 part whole · 5 marks

2025 JUN · Paper 02 · extract 2 of 2 · find part whole in this mark scheme extract

Question number	Scheme	Marks

Type MS · Trig equations that factorise

Answer 3 of 7 · Question 7 part b · 4 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

7 (a)	$(3x - 15 =)$ 30 **or** 330 **or** 390	M1
	$x = 15$ $x = 115$ $x = 135$	M1 A1 A1 [4]
(b)	$3\frac{\sin y}{\cos y} + 4\sin y = 0 \rightarrow \sin y\left(\frac{3}{\cos y} + 4\right) = 0$	M1
	$\sin y = 0$ and $\cos y = -\frac{3}{4}$ $\rightarrow$ $y =$	A1ft
	$y = -180, 0$	A1
	$y = 138.6, -138.6$	A1
	ALT	
	$3\tan y + 4\tan y \cos y = 0 \rightarrow \tan y\,(3 + 4\cos y) = 0$	M1
	$\tan y = 0$ and $\cos y = -\frac{3}{4}$ $\rightarrow$ $y =$	A1ft
	$y = -180, 0$	A1
	$y = 138.6, -138.6$	A1 [4]
(c)	$\cos\theta = 3\,(1 - \cos^2\theta) - 1$ $\rightarrow$ $3\cos^2\theta + \cos\theta - 2 = 0$	M1
	$(3\cos\theta - 2)(\cos\theta + 1)$	M1
	$\theta = -180$	A1
	$\theta = 48.2, -48.2$	A1 [4]
		Total 12 marks

Type MS · Trig equations that factorise

Answer 3 of 7 · Question 7 part b · 4 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For any of 30 **or** 330 **or** 390 May be implied by correct answers.
	M1 A1 A1	Solves a linear equation coming from attempt at use of inverse trigonometric function to obtain one value. e.g. solves $3x - 15 = "30"$ First A1 for any correct value, second A1 for all 3 correct values and no others in the range. Ignore values outside the range.
(b)	M1	Correctly replaces the identity for tan y and attempts to deal with $\sin y$. Allow for factorising. Condone dividing through by $\sin y$. Minimally acceptable attempt for factorisation is $A \sin y\left(\frac{B}{\cos y}+C\right)$
	A1ft	$\sin y = 0$ and $\cos y = -\frac{B}{C}$, follow through their B and C **only.**
	A1	From sine: Both values, ignore extra values out of range, A0 for extra values in range.
	A1	From cosine: Both values, ignore extra values out of range, A0 for extra values in range. y = awrt 138.6, awrt – 138.6
	ALT	
	M1	Correctly replaces the identity for tan y and attempts to deal with $\tan y$. Allow for factorising. Condone dividing through by $\tan y$. Minimally acceptable attempt for factorisation is $A\tan y\,(B + C\cos y)$
	A1ft	$\tan y = 0$ and $\cos y = -\frac{B}{C}$, follow through their B and C **only.**
	A1	From tangent: Both values, ignore extra values out of range, A0 for extra values in range.
	A1	From cosine: Both values, ignore extra values out of range, A0 for extra values in range. y = awrt 138.6, awrt – 138.6
(c)	M1	Correctly uses the identity for $\cos^2\theta$ and rearranges to get a 3TQ Minimally acceptable attempt is $\pm 3\cos^2\theta \pm \cos\theta \pm 2 = 0$
	M1	Solves a 3TQ to arrive at 2 distinct values for $\cos\theta$. See general guidance.
	A1	For – 180, ignore extra values out of range, A0 for extra values in range.
	A1	Both values, ignore extra values out of range, A0 for extra values in range. θ = awrt 48.2, awrt – 48.2

Type MS · Trig equations that factorise

Answer 3 of 7 · Question 7 part b · 4 marks

2022 JUN · Paper 02 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Trig equations that factorise

Answer 4 of 7 · Question 10 part c · 4 marks

2023 JAN · Paper 02 · find part c in this mark scheme extract

10 a (i)	$\sin 2\theta = \sin\theta\cos\theta + \cos\theta\sin\theta = 2\sin\theta\cos\theta$ *	M1 A1 cso (2)
a (ii)	$\cos 2\theta = \cos^2\theta - \sin^2\theta = \cos^2\theta - (1-\cos^2\theta)$	M1 M1
	$= 2\cos^2\theta - 1$ *	A1 cso (3)
b	$\sin 2\theta - \tan\theta = 2\sin\theta\cos\theta - \dfrac{\sin\theta}{\cos\theta}$	M1
	$\sin\theta\left(2\cos\theta - \dfrac{1}{\cos\theta}\right)$	M1
	$\sin\theta\left(\dfrac{2\cos^2\theta - 1}{\cos\theta}\right) = \tan\theta\cos 2\theta$ *	dM1 A1cso (4)
c	$\tan x = 0$ so $x = 180$	B1
	$\cos 2x = 0$ so $2x = 90, 270, 450, 630$	M1
	$x = 45, 135, 225, 315$	A1 A1 (4)
		Total 13 marks

Type MS · Trig equations that factorise

Answer 5 of 7 · Question 10 part c · 4 marks

2020 NOV · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

10(a)	$\cos(A+B)+\cos(A-B)=\cos A\cos B-\sin A\sin B+\cos A\cos B+\sin A\sin B$	M1
	$=2\cos A\cos B$ *	A1 cso (2)
(b)	Let $A+B=P,\quad A-B=Q \Rightarrow A=\frac{1}{2}(P+Q),\quad B=\frac{1}{2}(P-Q)$	M1
	As $\cos(A+B)+\cos(A-B)=2\cos A\cos B$	
	$\cos P+\cos Q=2\cos\frac{1}{2}(P+Q)\cos\frac{1}{2}(P-Q)$ *	M1A1cso (3)
ALT		
	Let $A=\frac{1}{2}(P+Q),\quad B=\frac{1}{2}(P-Q)\Rightarrow A+B=P,\ A-B=Q$	M1
	As $2\cos A\cos B=\cos(A+B)+\cos(A-B)$	
	$2\cos\frac{1}{2}(P+Q)\cos\frac{1}{2}(P-Q)=\cos P+\cos Q$ *	M1A1cso
(c)	$\cos 5\theta+\cos 7\theta=2\cos 6\theta\cos\theta=0$	M1
	$\cos 6\theta=0\Rightarrow 6\theta=\frac{\pi}{2},\frac{3\pi}{2},\frac{5\pi}{2}\Rightarrow\theta=\frac{\pi}{12},\frac{\pi}{4}\ \left(\text{or}\ \frac{3\pi}{12}\right),\frac{5\pi}{12}$	A1A1
	$\cos\theta=0\Rightarrow\theta=\frac{\pi}{2}$	A1 (4)
(d)	$\cos 8x+2\cos 6x+\cos 4x=(\cos 8x+\cos 6x)+(\cos 6x+\cos 4x)$	
	$=2\cos 7x\cos x+2\cos 5x\cos x$	M1
	$=2\cos x(\cos 7x+\cos 5x)=2\cos x\times 2(\cos 6x\cos x),\ =4\cos 6x\cos^2 x$ *	dM1,A1cso (3)
ALT 1	$\cos 8x+2\cos 6x+\cos 4x=(\cos 8x+\cos 4x)+2\cos 6x$	
	$=2\cos 6x\cos 2x+2\cos 6x$	M1
	$=2\cos 6x(\cos 2x+1)=2\cos 6x(2\cos^2 x-1+1)=4\cos 6x\cos^2 x$ *	dM1A1cso (3)
ALT 2	Working from right to left	
	$4\cos 6x\cos^2 x=4\cos 6x\times\frac{1}{2}(\cos 2x+1)=2\cos 6x\cos 2x+2\cos 6x$	M1
	$=\cos 8x+\cos 4x+2\cos 6x$ *	dM1A1cso (3)

Type MS · Trig equations that factorise

Answer 5 of 7 · Question 10 part c · 4 marks

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(e)	$\int_0^{\frac{\pi}{3}} \cos 6x \cos^2 x \, dx = \frac{1}{4}\int_0^{\frac{\pi}{3}} (\cos 8x + 2\cos 6x + \cos 4x) \, dx$	M1	
	$= \frac{1}{4}\left[\frac{1}{8}\sin 8x + \frac{1}{3}\sin 6x + \frac{1}{4}\sin 4x\right]_0^{\frac{\pi}{3}}$	A1	
	$= \frac{1}{4}\left[\frac{1}{8}\sin\frac{8\pi}{3} + \frac{1}{3}\sin 2\pi + \frac{1}{4}\sin\frac{4\pi}{3} - 0\right] = -\frac{\sqrt{3}}{64}$ (-0.02706... scores M1A0)	dM1A1	(4)
			[16]
ALT	For the first M mark Award M1 only when the integrand has been changed (by a valid method) to a function which can be integrated $\int_0^{\frac{\pi}{3}} \cos 6x \cos^2 x \, dx = \int_0^{\frac{\pi}{3}} \cos 6x \times \frac{1}{2}(\cos 2x + 1) \, dx$		
	$= \frac{1}{2}\int_0^{\frac{\pi}{3}} (\cos 6x \times \cos 2x + \cos 6x) \, dx = \frac{1}{2}\int_0^{\frac{\pi}{3}} \frac{1}{2}(\cos 8x + \cos 4x + 2\cos 6x) \, dx$	M1	
	Rest as above.		

(a)	
M1	Use the standard formulae. This is a "show that" question so these formulae must be written out in full. Both sides of the result must be seen although the working may appear between them as seen in the scheme.
A1cso	Final **given** result. Do not award if the expansions shown are not in the correct order (this suggests incorrect signs in the formulae).
(b)	
M1	Use $A+B=P,\quad A-B=Q$ to obtain A and B in terms of P and Q
M1	Substitute in the result from (a) to obtain an identity in P and Q only
A1cso	Correct result reached with no errors seen
ALT	Working right to left: Notes similar to above
(c)	
M1	Use the result from (b) to show that $(2)\cos 6\theta \cos(\pm\theta) = 0$
A1	Obtain one correct solution of $\cos 6\theta = 0$. Allow if in decimal form but must be radians.
A1	Two further correct solutions **and no more within the range.** If any of the 3 solutions of $\cos 6\theta = 0$ is not exact do not award this mark.
A1	State the solution of $\cos(\pm\theta) = 0$. Must be exact unless this is penalised above.
NB	If answers are given in degrees deduct 2A marks from any that would otherwise have been given. (Answers in degrees but then changed to radians are acceptable – mark the radian answers.) Ignore extra answers outside the stated range – any within are incorrect.
(d)	
M1	Use the result from (a) or (b) once. Allow if one or both "2"s are missing
dM1	Use the result from (a) or (b) again. Must include both "2"s this time. Depends on previous M mark.
A1cso	Reach the **given** result with no errors seen

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Answer 5 of 7 · Question 10 part c · 4 marks

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ALT 1	
M1	Use the result from (a) or (b). Allow if the "2" is missing.
dM1	Factorise and use the *correct* double angle formula on cos2x Depends on the previous M mark
A1cso	Reach the **given** result with no errors seen
ALT 2	
M1	Use the *correct* half angle formula on $\cos^2 x$
dM1A1	M1: Use the result from (a) or (b)
cso	A1: reach the **given** result with no errors seen
(e)M1	Obtain a function which can be integrated **either** by using the **given** result from (d) **OR**. deriving the same result. Allow if 1/4 is missing. (Integration by Parts – send to review)
A1	Correct integration (must have included the ¼) Limits not needed for these 2 marks.
dM1	Correct use of the given limits. All sines are 0 at the lower limit so these need not be shown
A1	Correct final answer which must be **exact and stated as a single fraction**.

Type MS · Trig equations that factorise

Answer 6 of 7 · Question 8 part c · 6 marks

2020 JAN · Paper 02 (regional R) · extract 1 of 3 · find part c in this mark scheme extract

8 (a)	$k = 2$	B1 [1]
(b)	$2+3\cos A-\sin A-3\sin 2A-2\cos^2 A=3\cos A-\sin A-6\sin A\cos A+2\sin^2 A$ $3\cos A-\sin A-6\sin A\cos A+2\sin^2 A=3\cos A(1-2\sin A)-\sin A(1-2\sin A)$ $3\cos A(1-2\sin A)-\sin A(1-2\sin A)=(1-2\sin A)(3\cos A-\sin A)$ $\Rightarrow p=3,\ q=1,\ r=2$	B1 M1 A1 [3]
	ALT	
	$2+3\cos A-\sin A-3\sin 2A-2\cos^2 A=3\cos A-\sin A-6\sin A\cos A+2\sin^2 A$ $(p\cos A-\sin A)(q-r\sin A)=pq\cos A-q\sin A-pr\cos A\sin A+r\sin^2 A$ $pq\cos A-q\sin A-pr\cos A\sin A+r\sin^2 A$ $\Rightarrow p=3,\ q=1,\ r=2$	{B1} {M1} {A1} [3]
(c)	$(1-2\sin 2\theta)(3\cos 2\theta-\sin 2\theta)=0$ $\Rightarrow 3\cos 2\theta-\sin 2\theta=0\Rightarrow\tan 2\theta=3$ $\Rightarrow 1-2\sin 2\theta=0\Rightarrow\sin 2\theta=\frac{1}{2}$ $\tan 2\theta=3\Rightarrow 2\theta=1.2490...,\ 4.3906...\Rightarrow\theta=0.625,\ 2.20$ $\sin 2\theta=\frac{1}{2}\Rightarrow 2\theta=0.5235...,\ 2.6179...\Rightarrow\theta=0.262,\ 1.31\ \left[\frac{\pi}{12},\frac{5\pi}{12}\right]$	B1 B1 M1A1 M1A1 [6]
		Total 10 marks

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Answer 6 of 7 · Question 8 part c · 6 marks

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(c)	
B1	$\tan 2\theta = 3$
B1	$\sin 2\theta = \frac{1}{2}$
M1	$2\theta = 1.24(90...,)\ 4.39(06...)$
A1	$\theta = 0.625,\ 2.2(0)$
M1	$2\theta = 0.5235...,\ 2.61(79...)$ allow $\left[\frac{\pi}{6}, \frac{5\pi}{6}\right]$
A1	$\theta = 0.262,\ 1.31$ allow $\left[\frac{\pi}{12}, \frac{5\pi}{12}\right]$

Type MS · Trig equations that factorise

Answer 6 of 7 · Question 8 part c · 6 marks

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Question number	Scheme	Marks

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Question	Scheme	Marks
10(a)	$\cos(A-B)=\cos A\cos B+\sin A\sin B$ $\cos(A+B)=\cos A\cos B-\sin A\sin B$ $\cos(A-B)-\cos(A+B)=\sin A\sin B-(-\sin A\sin B)=2\sin A\sin B$ *	M1A1cso [2]
(b)	$[A-B=5\theta,\ A+B=9\theta\Rightarrow A=7\theta,\ B=2\theta]$ $\cos 5\theta-\cos 9\theta=2\sin 7\theta\sin 2\theta$ *	B1 cso [1]
(c)	$\sqrt{3}\sin 7\theta=2\sin 7\theta\sin 2\theta\Rightarrow 0=2\sin 7\theta\sin 2\theta-\sqrt{3}\sin 7\theta$ $0=\sin 7\theta\left(2\sin 2\theta-\sqrt{3}\right)\Rightarrow \sin 7\theta=0,\ 2\sin 2\theta-\sqrt{3}=0$ $\sin 7\theta=0\Rightarrow 7\theta=0,\ \pi,\ 2\pi\Rightarrow\theta=\frac{\pi}{7},\ \frac{2\pi}{7}$ $2\sin 2\theta-\sqrt{3}=0\Rightarrow\sin 2\theta=\frac{\sqrt{3}}{2}\Rightarrow 2\theta=\frac{\pi}{3},\ \frac{2\pi}{3}\Rightarrow\theta=\frac{\pi}{6},\ \frac{\pi}{3}$	M1A1 M1 M1A1 M1A1 [7]
(d)	$\tan 2x=\frac{\sin 2x}{\cos 2x}\Rightarrow\tan 2x\cos 2x=\sin 2x$ $\int_0^{\frac{\pi}{7}}8\sin 7x\sin 2x\,dx=\left[\int_0^{\frac{\pi}{7}}4\times(2\sin 7x\sin 2x)\,dx\right]$ $\int_0^{\frac{\pi}{7}}4(\cos 5x-\cos 9x)\,dx=4\left[\frac{\sin 5x}{5}-\frac{\sin 9x}{9}\right]_0^{\frac{\pi}{7}}$ $4\left[\frac{\sin 5x}{5}-\frac{\sin 9x}{9}\right]_0^{\frac{\pi}{7}}=4\left[\left(\frac{\sin 5\times\frac{\pi}{7}}{5}-\frac{\sin 9\times\frac{\pi}{7}}{9}\right)-\left(\frac{\sin 0}{5}-\frac{\sin 0}{9}\right)\right]$ # $=0.9729...\approx 0.973$	M1 M1 M1M1 M1A1 [6]
		Total 16 marks

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Question	Notes	Marks
10(a)	From the formula sheet $\cos(A-B)=\cos A\cos B+\sin A\sin B$ $\cos(A+B)=\cos A\cos B-\sin A\sin B$ Subtracts the two equations to give: $\cos(A-B)-\cos(A+B)=\sin A\sin B-(-\sin A\sin B)$	M1
	For the correct identity as shown with no errors, $\cos(A-B)-\cos(A+B)=2\sin A\sin B$ *	A1 cso [2]
(b)	For finding the value of A and the value of B $A-B=5\theta,\quad A+B=9\theta$ $\Rightarrow A=7\theta,\ B=2\theta$ Or as a minimum: $\cos(7\theta-2\theta)-\cos(7\theta+2\theta)=2\sin 7\theta\sin 2\theta$ *	B1cso [1]
(c)	Sets $\sqrt{3}\sin 7\theta=2\sin 7\theta\sin 2\theta$	M1
	Achieves the correct equation allow the terms in any order. $0=2\sin 7\theta\sin 2\theta-\sqrt{3}\sin 7\theta$	A1
	Factorises their equation $0=\sin 7\theta\left(2\sin 2\theta-\sqrt{3}\right)\Rightarrow \sin 7\theta=0,\ 2\sin 2\theta-\sqrt{3}=0$	M1
	For finding at least **one** correct value for θ using $\sin 7\theta=0$ $\sin 7\theta=0\Rightarrow 7\theta=0,\ \pi,\ 2\pi\Rightarrow\theta=\frac{\pi}{7},\ \frac{2\pi}{7}$	M1
	For both correct values $\theta=\frac{\pi}{7},\ \frac{2\pi}{7}$	A1
	For finding **one** correct value of θ using $2\sin 2\theta-\sqrt{3}=0$ $2\sin 2\theta-\sqrt{3}=0\Rightarrow\sin 2\theta=\frac{\sqrt{3}}{2}\Rightarrow 2\theta=\frac{\pi}{3},\ \frac{2\pi}{3}\Rightarrow\theta=\frac{\pi}{6},\ \frac{\pi}{3}$	M1
	For both correct values $\theta=\frac{\pi}{6},\ \frac{\pi}{3}$	A1 [7]
(d)	Uses the identity for $\tan 2x=\frac{\sin 2x}{\cos 2x}\Rightarrow\tan 2x\cos 2x=\sin 2x$	M1
	Substitutes the above into $8\sin 7x\cos 2x\tan 2x$ to give $\int_0^{\frac{\pi}{7}}8\sin 7x\sin 2x\,\mathrm{d}x=\left[\int_0^{\frac{\pi}{7}}4\times(2\sin 7x\sin 2x)\,\mathrm{d}x\right]$ Ignore integral sign and limits for this mark.	M1
	For substituting $\cos 5x-\cos 9x$ for $2\sin 7x\sin 2x$ to give $\int_0^{\frac{\pi}{7}}4(\cos 5x-\cos 9x)\,\mathrm{d}x$ Ignore integral sign and limits for this mark.	M1

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	Integrates $\int_0^{\frac{\pi}{7}} 4(\cos 5x - \cos 9x)\,dx = 4\left[\frac{\sin 5x}{5} - \frac{\sin 9x}{9}\right]_0^{\frac{\pi}{7}}$ Ignore limits for this mark. As a minimum they must obtain: $\left(\pm\frac{\sin 5x}{5} \pm \frac{\sin 9x}{9}\right)$ for the integration.	M1
	Substitutes the limits the correct way around $4\left[\frac{\sin 5x}{5} - \frac{\sin 9x}{9}\right]_0^{\frac{\pi}{7}} = 4\left[\left(\frac{\sin 5\times\frac{\pi}{7}}{5} - \frac{\sin 9\times\frac{\pi}{7}}{9}\right) - \left(\frac{\sin 0}{5} - \frac{\sin 0}{9}\right)\right]$ $= (0.9729...)$	M1
	For the correct value of $\int_0^{\frac{\pi}{7}} 8\sin 7x\cos 2x\tan 2x\,dx = 0.973$	A1 [6]
		Total 16 marks

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